Universidad Rafael Landívar

Facultad de Ingeniería

Ingeniería en Informática y de Sistemas

Ciencia de Datos

Sección No. 02

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Fase 1

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Guatemala, 23 de septiembre de 2026

Evidencias

Figura 1. Estructura inicial del proyecto de la Red Metropolitana

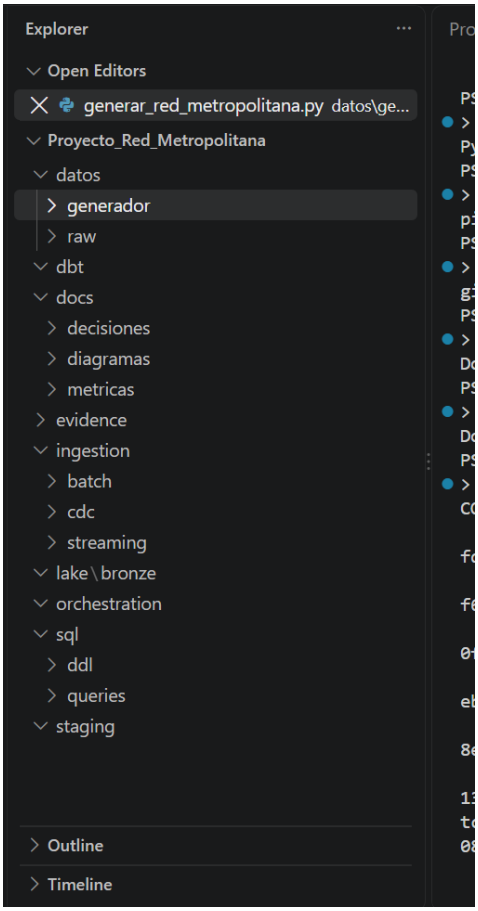


Figura 2. Infraestructura de PostgreSQL y Kafka ejecutándose en Docker

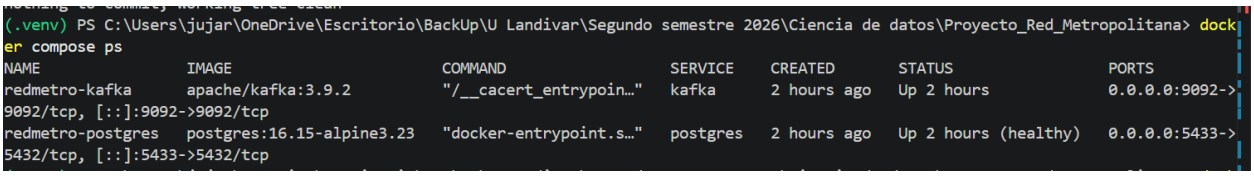


Figura 3. Ingesta Batch de los catálogos hacia la capa Bronze

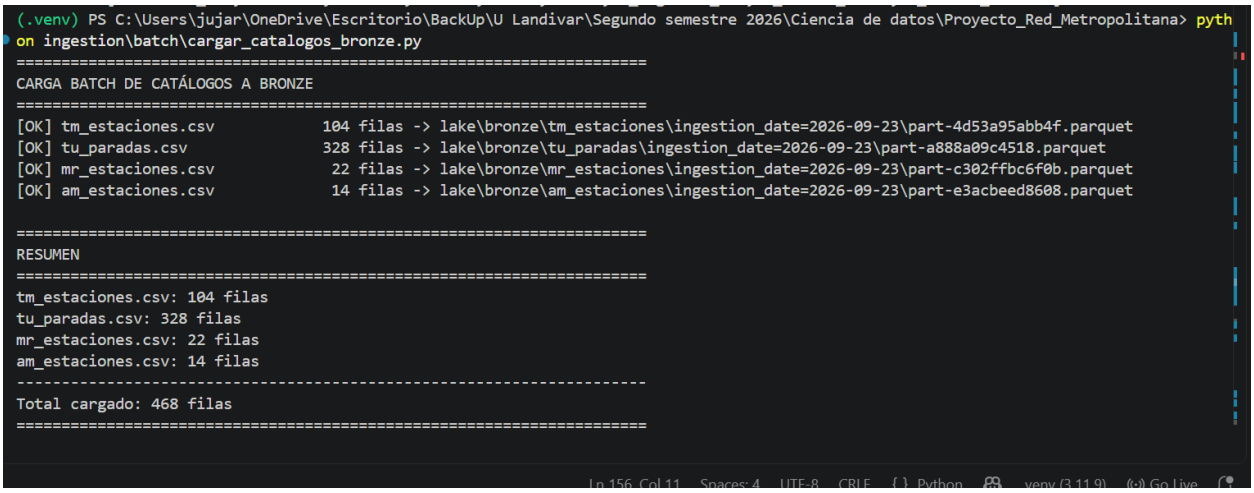


Figura 4. Catálogos almacenados en Bronze y particionados por fecha de ingesta

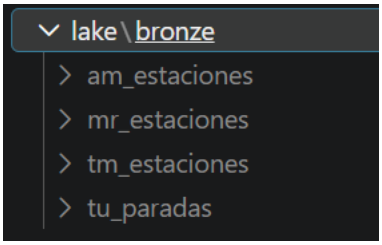


Figura 5. Ingesta Batch de los viajes de MetroRiel hacia Bronze

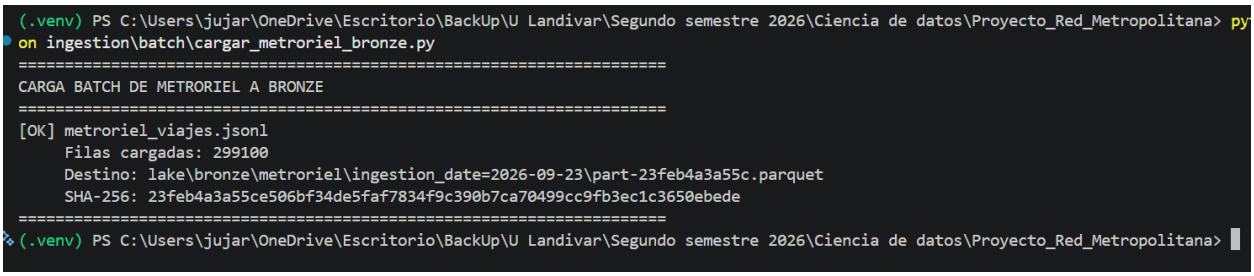


Figura 6. Publicación de validaciones de Transmetro mediante Kafka

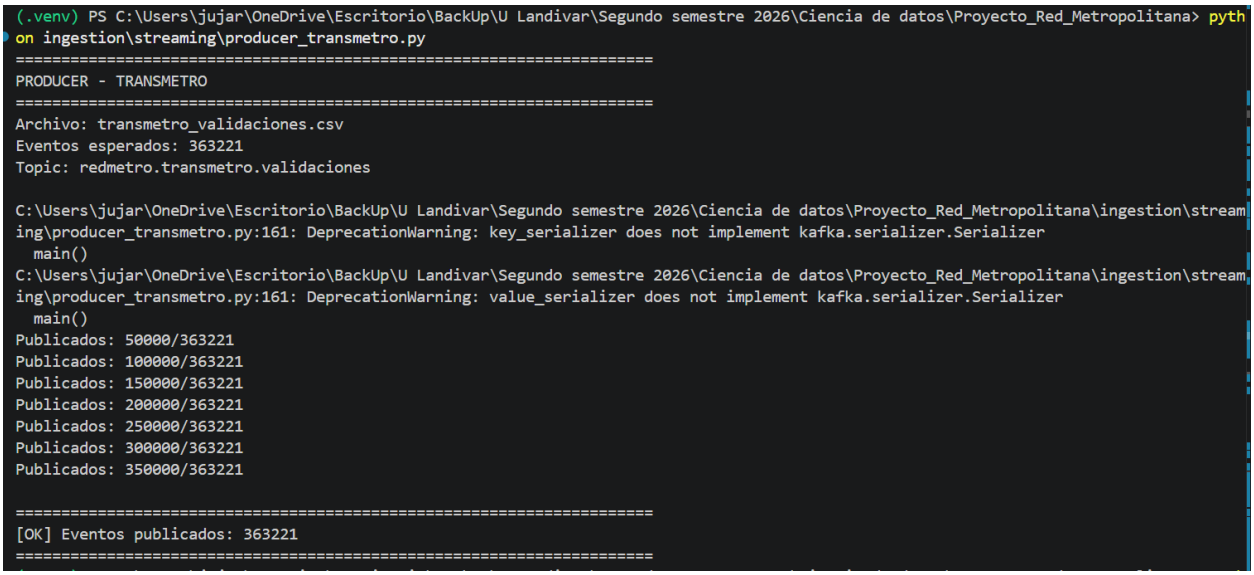


Figura 7. Consumo de eventos de Transmetro y almacenamiento en Bronze

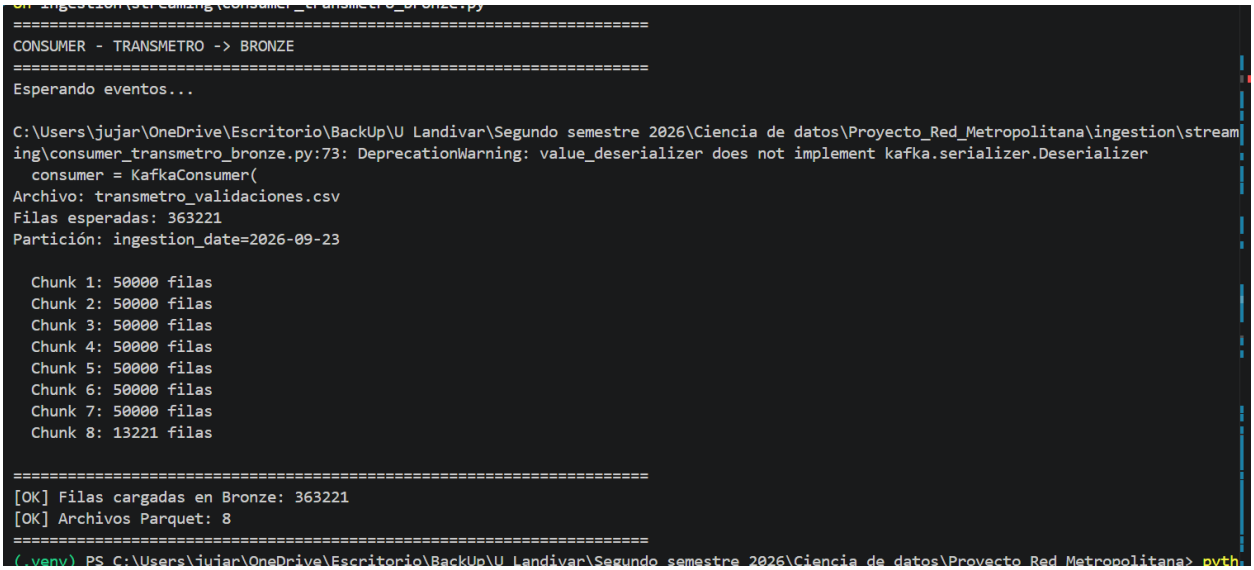


Figura 8. Ingesta streaming de abordajes de Aerómetro hacia Bronze

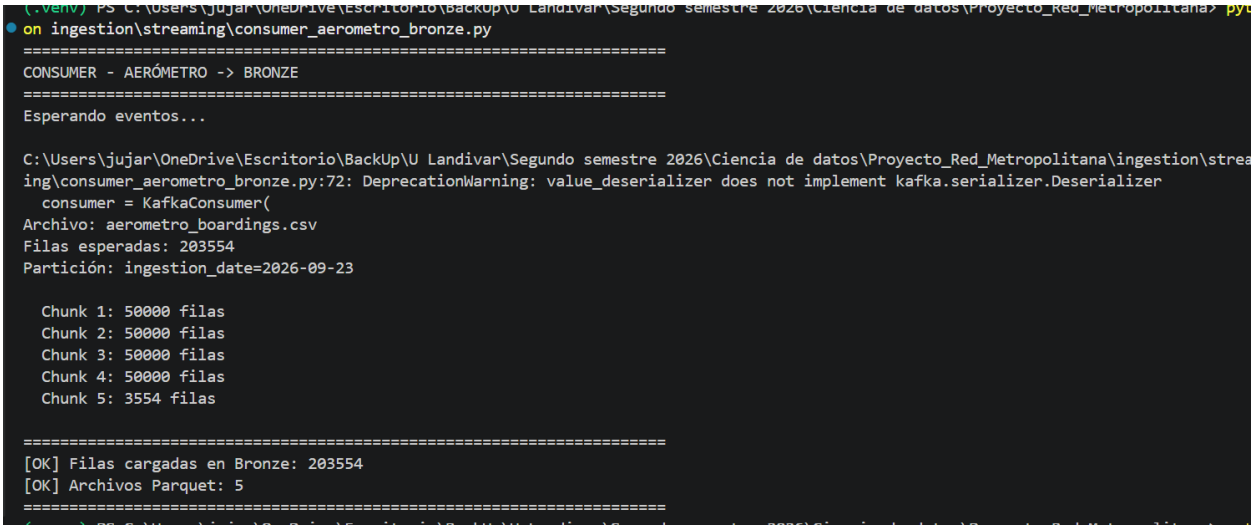


Figura 9. Ingesta streaming de transacciones de Transurbano hacia Bronze

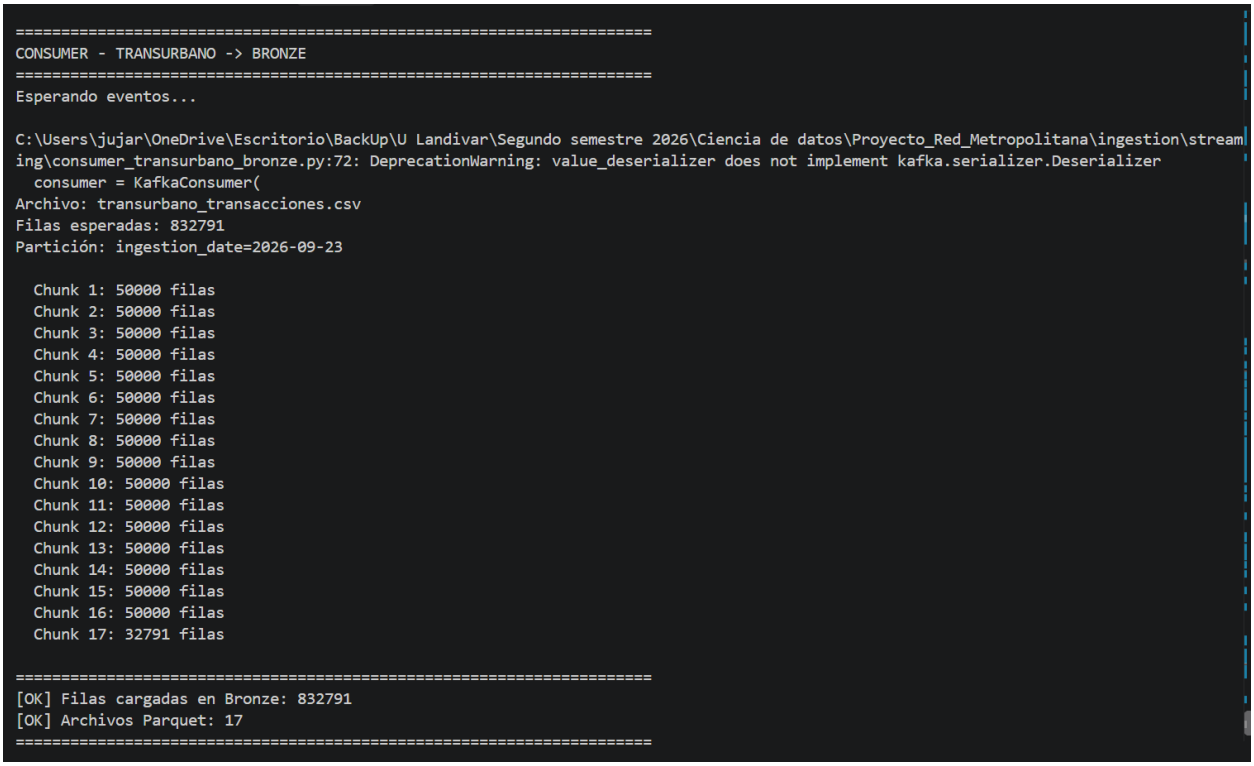
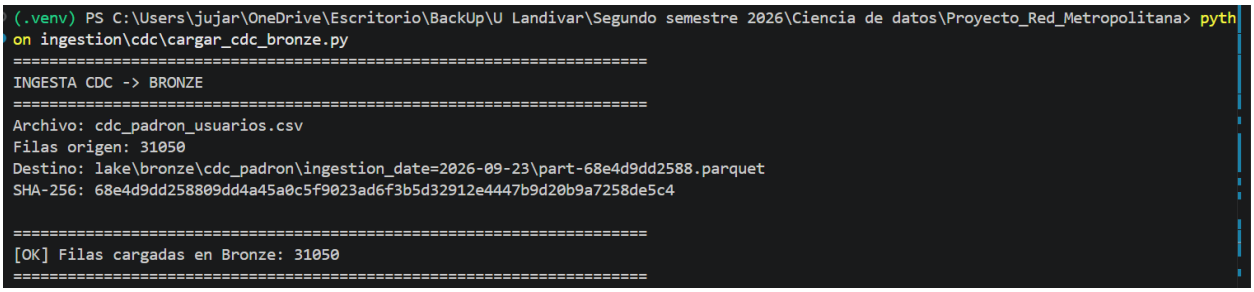


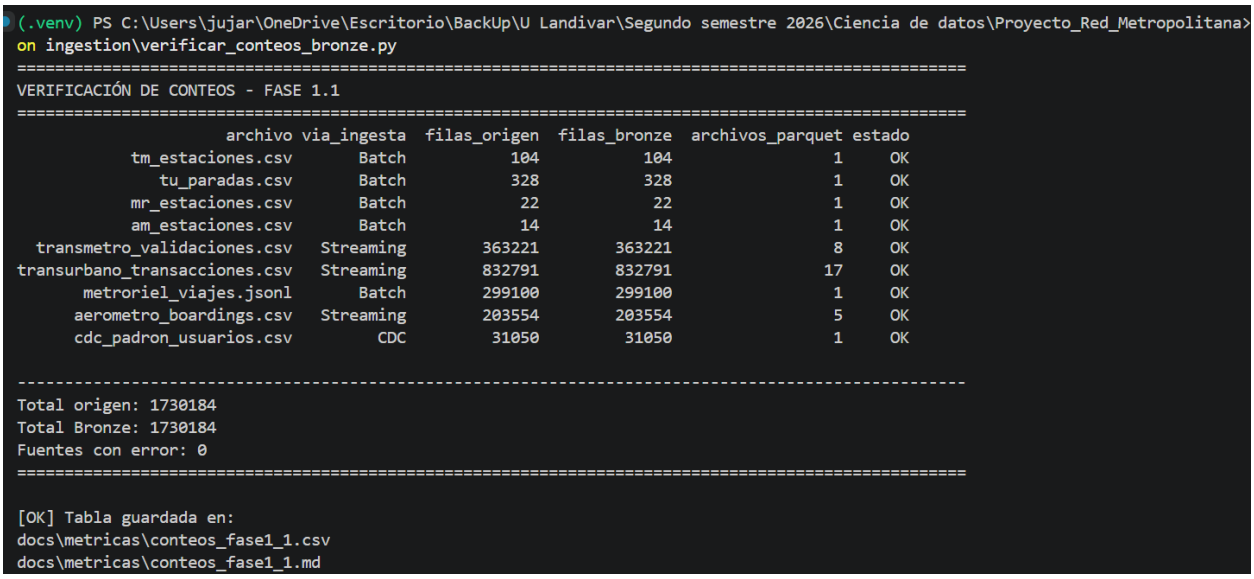
Figura 10. Ingesta del archivo CDC del padrón hacia Bronze



```
(.venv) PS C:\Users\jujar\OneDrive\Escritorio\BackUp\U Landivar\Segundo semestre 2026\Ciencia de datos\Proyecto_Red_Metropolitana> pyth
on ingestion\cdc\cargar_cdc_bronze.py
======================================================================
INGESTA CDC -> BRONZE
======================================================================
Archivo: cdc_padron_usuarios.csv
Filas origen: 31050
Destino: lake\bronze\cdc_padron\ingestion_date=2026-09-23\part-68e4d9dd2588.parquet
SHA-256: 68e4d9dd258809dd4a45a0c5f9023ad6f3b5d32912e4447b9d20b9a7258de5c4

======================================================================
[OK] Filas cargadas en Bronze: 31050
======================================================================
```

Figura 11. Verificación de conteos de las fuentes en la capa Bronze



```
(.venv) PS C:\Users\jujar\OneDrive\Escritorio\BackUp\U Landivar\Segundo semestre 2026\Ciencia de datos\Proyecto_Red_Metropolitana>
on ingestion\verificar_conteos_bronze.py
================================================================================================
VERIFICACIÓN DE CONTEOS - FASE 1.1
================================================================================================
                    archivo via_ingesta  filas_origen  filas_bronze  archivos_parquet estado
          tm_estaciones.csv       Batch           104           104                 1     OK
             tu_paradas.csv       Batch           328           328                 1     OK
          mr_estaciones.csv       Batch            22            22                 1     OK
          am_estaciones.csv       Batch            14            14                 1     OK
  transmetro_validaciones.csv   Streaming        363221        363221                 8     OK
transmetro_transacciones.csv   Streaming        832791        832791                17     OK
       metroriel_viajes.jsonl       Batch        299100        299100                 1     OK
      aerometro_boardings.csv   Streaming        203554        203554                 5     OK
      cdc_padron_usuarios.csv         CDC         31050         31050                 1     OK

------------------------------------------------------------------------------------------------
Total origen: 1730184
Total Bronze: 1730184
Fuentes con error: 0
================================================================================================

[OK] Tabla guardada en:
docs\metricas\conteos_fase1_1.csv
docs\metricas\conteos_fase1_1.md
```

Figura 12. Procesamiento del CDC y construcción del padrón vigente de Transmetro

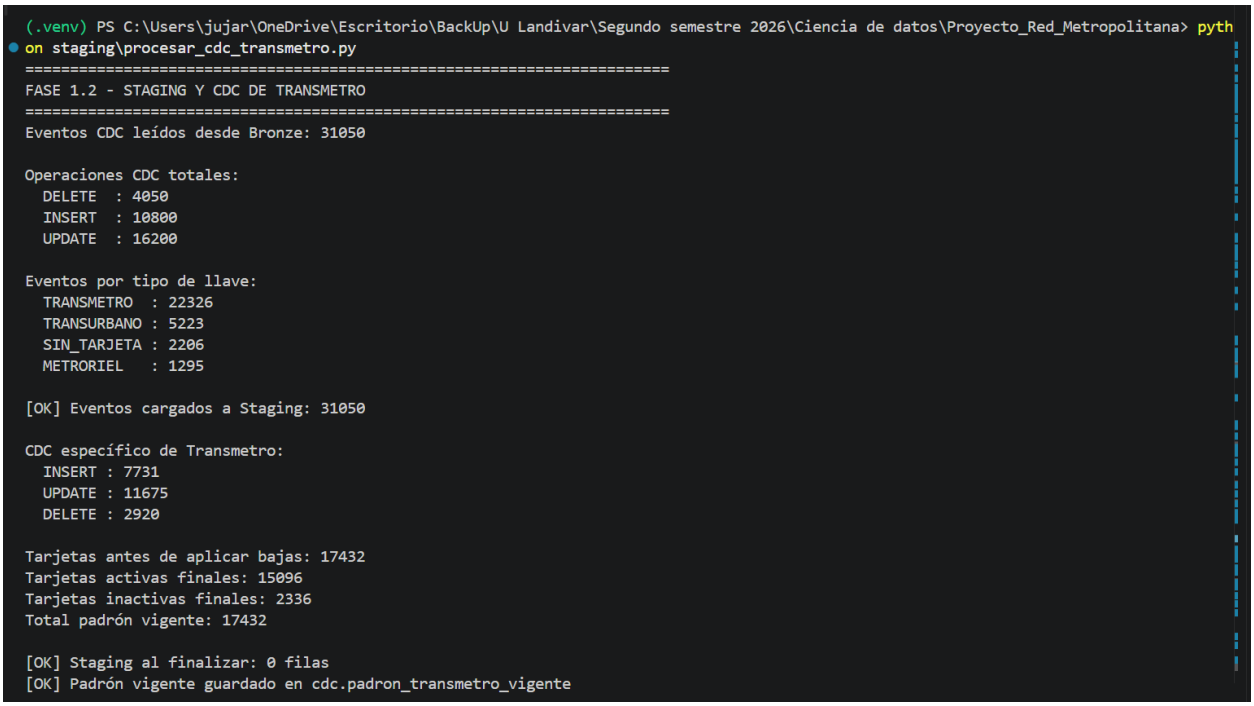


Figura 13. Conteo de tarjetas activas e inactivas después de aplicar el CDC

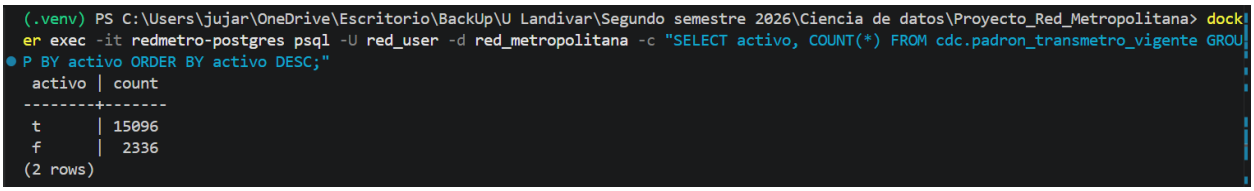


Figura 14. Construcción de catálogos mínimos de usuarios por operador

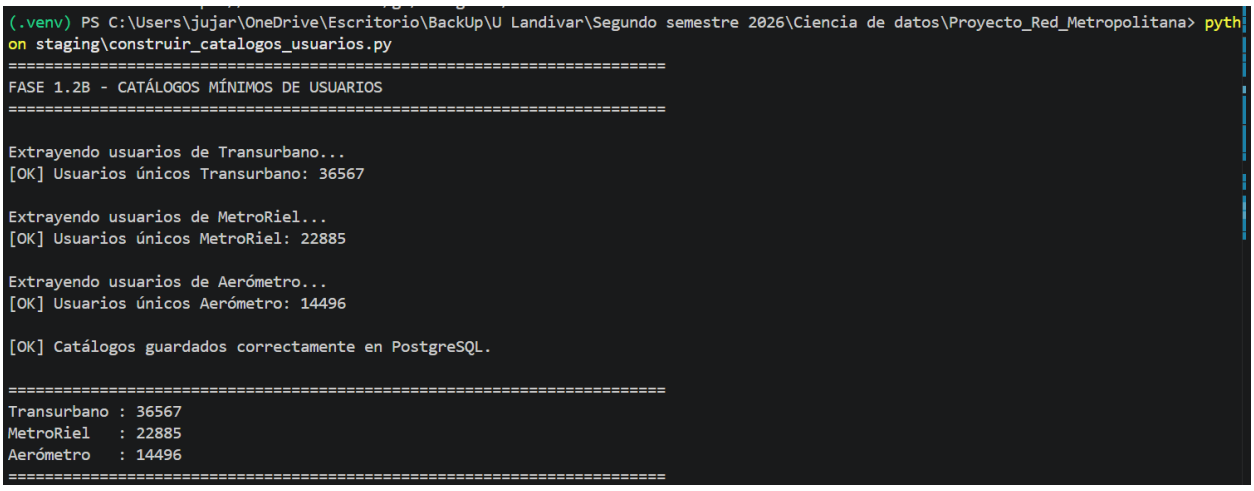
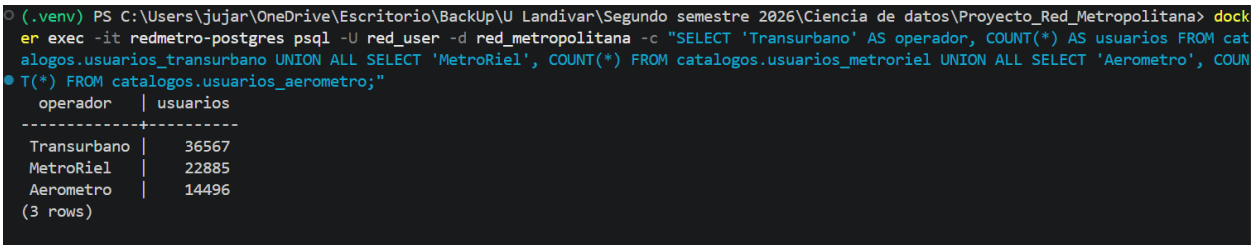
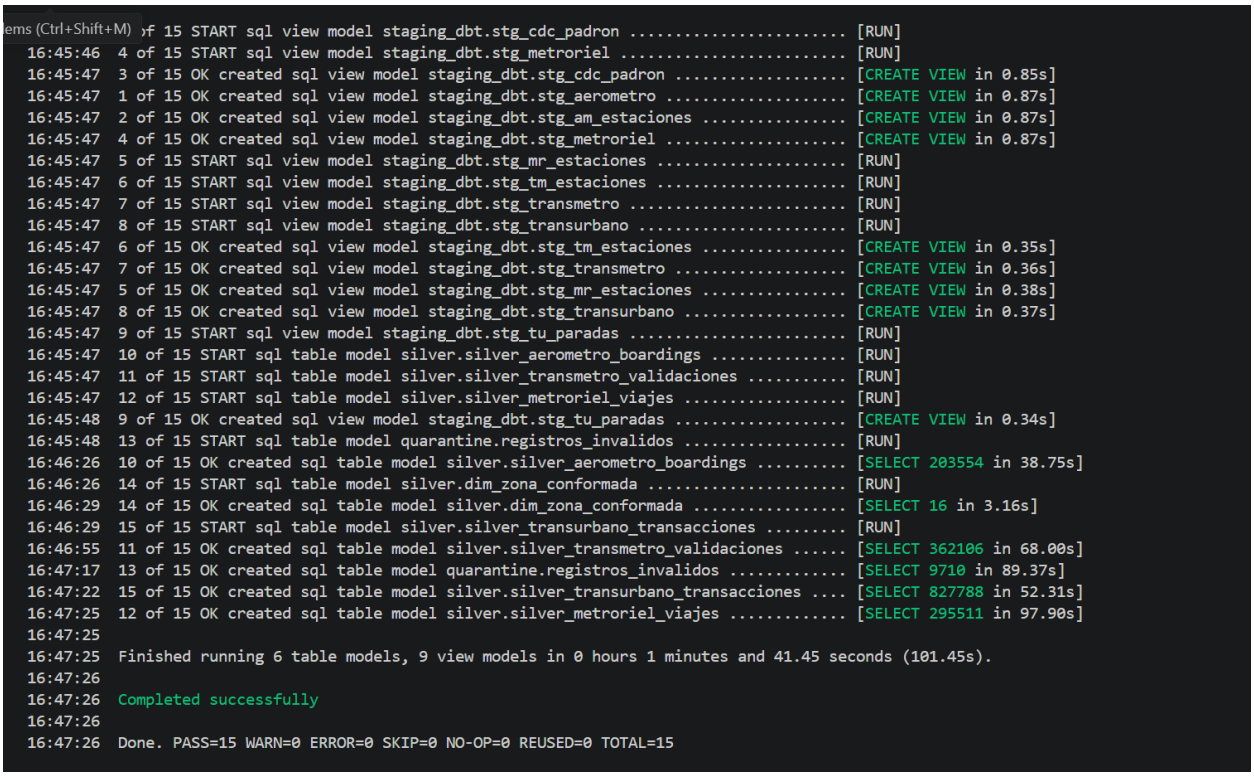


Figura 15. Verificación de usuarios únicos en los catálogos mínimos



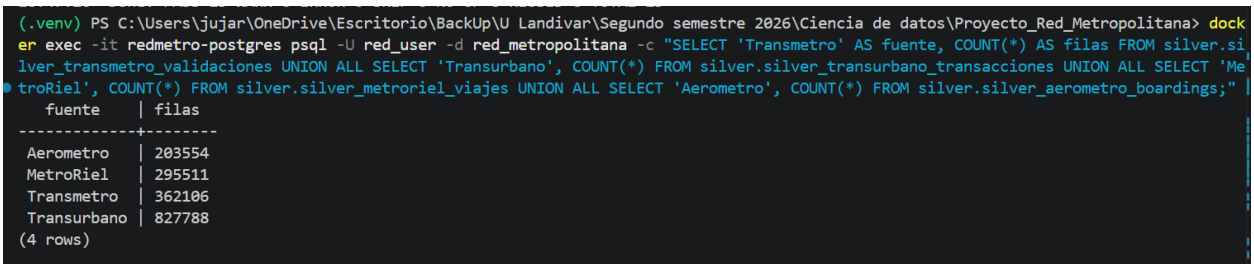
```
(.venv) PS C:\Users\jujar\OneDrive\Escritorio\BackUp\U Landivar\Segundo semestre 2026\Ciencia de datos\Proyecto_Red_Metropolitana> docker exec -it redmetro-postgres psql -U red_user -d red_metropolitana -c "SELECT 'Transurbano' AS operador, COUNT(*) AS usuarios FROM catalogos.usuarios_transurbano UNION ALL SELECT 'MetroRiel', COUNT(*) FROM catalogos.usuarios_metroriel UNION ALL SELECT 'Aerometro', COUNT(*) FROM catalogos.usuarios_aerometro;"
  operador   | usuarios
-------------+----------
 Transurbano |    36567
 MetroRiel   |    22885
 Aerometro   |    14496
(3 rows)
```

Figura 16. Ejecución de modelos dbt para las capas Silver y cuarentena



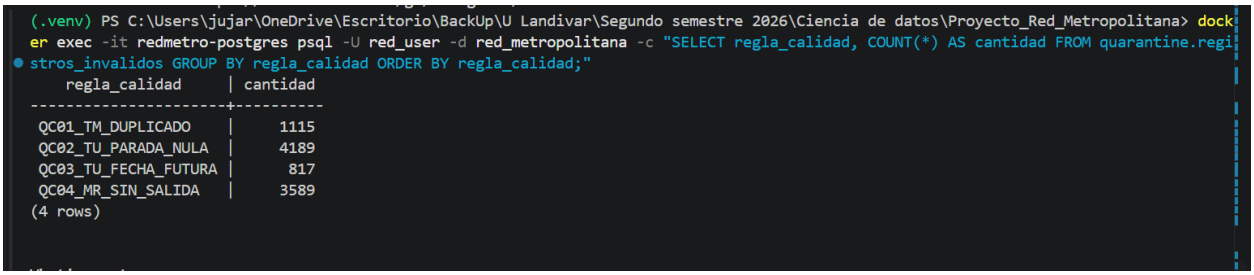
```
ems (Ctrl+Shift+M) of 15 START sql view model staging_dbt.stg_cdc_padron ..... [RUN]
16:45:46  4 of 15 START sql view model staging_dbt.stg_metroriel ..... [RUN]
16:45:47  3 of 15 OK created sql view model staging_dbt.stg_cdc_padron ..... [CREATE VIEW in 0.85s]
16:45:47  1 of 15 OK created sql view model staging_dbt.stg_aerometro ..... [CREATE VIEW in 0.87s]
16:45:47  2 of 15 OK created sql view model staging_dbt.stg_am_estaciones ..... [CREATE VIEW in 0.87s]
16:45:47  4 of 15 OK created sql view model staging_dbt.stg_metroriel ..... [CREATE VIEW in 0.87s]
16:45:47  5 of 15 START sql view model staging_dbt.stg_mr_estaciones ..... [RUN]
16:45:47  6 of 15 START sql view model staging_dbt.stg_tm_estaciones ..... [RUN]
16:45:47  7 of 15 START sql view model staging_dbt.stg_transmetro ..... [RUN]
16:45:47  8 of 15 START sql view model staging_dbt.stg_transurbano ..... [RUN]
16:45:47  6 of 15 OK created sql view model staging_dbt.stg_tm_estaciones ..... [CREATE VIEW in 0.35s]
16:45:47  7 of 15 OK created sql view model staging_dbt.stg_transmetro ..... [CREATE VIEW in 0.36s]
16:45:47  5 of 15 OK created sql view model staging_dbt.stg_mr_estaciones ..... [CREATE VIEW in 0.38s]
16:45:47  8 of 15 OK created sql view model staging_dbt.stg_transurbano ..... [CREATE VIEW in 0.37s]
16:45:47  9 of 15 START sql view model staging_dbt.stg_tu_paradas ..... [RUN]
16:45:47  10 of 15 START sql table model silver.silver_aerometro_boardings ..... [RUN]
16:45:47  11 of 15 START sql table model silver.silver_transmetro_validaciones ..... [RUN]
16:45:47  12 of 15 START sql table model silver.silver_metroriel_viajes ..... [RUN]
16:45:48  9 of 15 OK created sql view model staging_dbt.stg_tu_paradas ..... [CREATE VIEW in 0.34s]
16:45:48  13 of 15 START sql table model quarantine.registros_invalidos ..... [RUN]
16:46:26  10 of 15 OK created sql table model silver.silver_aerometro_boardings ..... [SELECT 203554 in 38.75s]
16:46:26  14 of 15 START sql table model silver.dim_zona_conformada ..... [RUN]
16:46:29  14 of 15 OK created sql table model silver.dim_zona_conformada ..... [SELECT 16 in 3.16s]
16:46:29  15 of 15 START sql table model silver.silver_transurbano_transacciones ..... [RUN]
16:46:55  11 of 15 OK created sql table model silver.silver_transmetro_validaciones ..... [SELECT 362106 in 68.00s]
16:47:17  13 of 15 OK created sql table model quarantine.registros_invalidos ..... [SELECT 9710 in 89.37s]
16:47:22  15 of 15 OK created sql table model silver.silver_transurbano_transacciones .... [SELECT 827788 in 52.31s]
16:47:25  12 of 15 OK created sql table model silver.silver_metroriel_viajes ..... [SELECT 295511 in 97.90s]
16:47:25
16:47:25  Finished running 6 table models, 9 view models in 0 hours 1 minutes and 41.45 seconds (101.45s).
16:47:26
16:47:26  Completed successfully
16:47:26
16:47:26  Done. PASS=15 WARN=0 ERROR=0 SKIP=0 NO-OP=0 REUSED=0 TOTAL=15
```

Figura 17. Conteo de registros válidos por operador en Silver



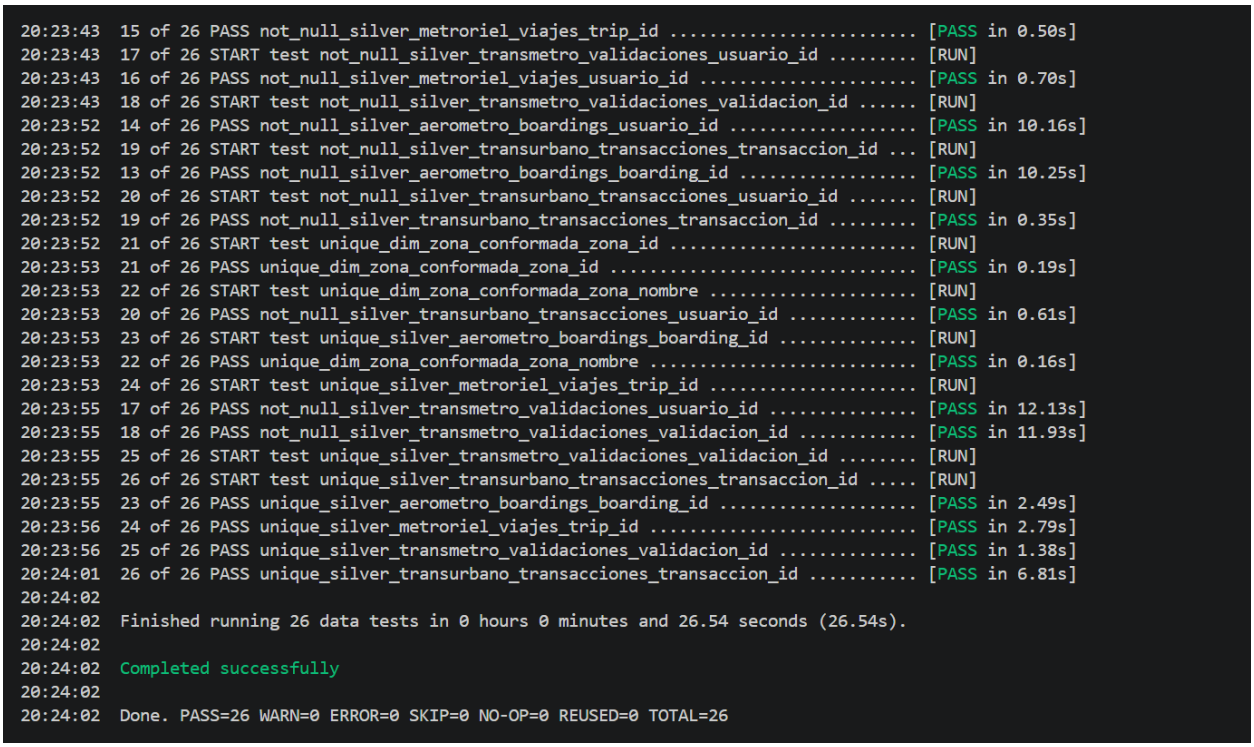
```
(.venv) PS C:\Users\jujar\OneDrive\Escritorio\BackUp\U Landivar\Segundo semestre 2026\Ciencia de datos\Proyecto_Red_Metropolitana> docker exec -it redmetro-postgres psql -U red_user -d red_metropolitana -c "SELECT 'Transmetro' AS fuente, COUNT(*) AS filas FROM silver.silver_transmetro_validaciones UNION ALL SELECT 'Transurbano', COUNT(*) FROM silver.silver_transurbano_transacciones UNION ALL SELECT 'MetroRiel', COUNT(*) FROM silver.silver_metroriel_viajes UNION ALL SELECT 'Aerometro', COUNT(*) FROM silver.silver_aerometro_boardings;"
   fuente    | filas
-------------+--------
 Aerometro   | 203554
 MetroRiel   | 295511
 Transmetro  | 362106
 Transurbano | 827788
(4 rows)
```

Figura 18. Registros enviados a cuarentena por regla de calidad



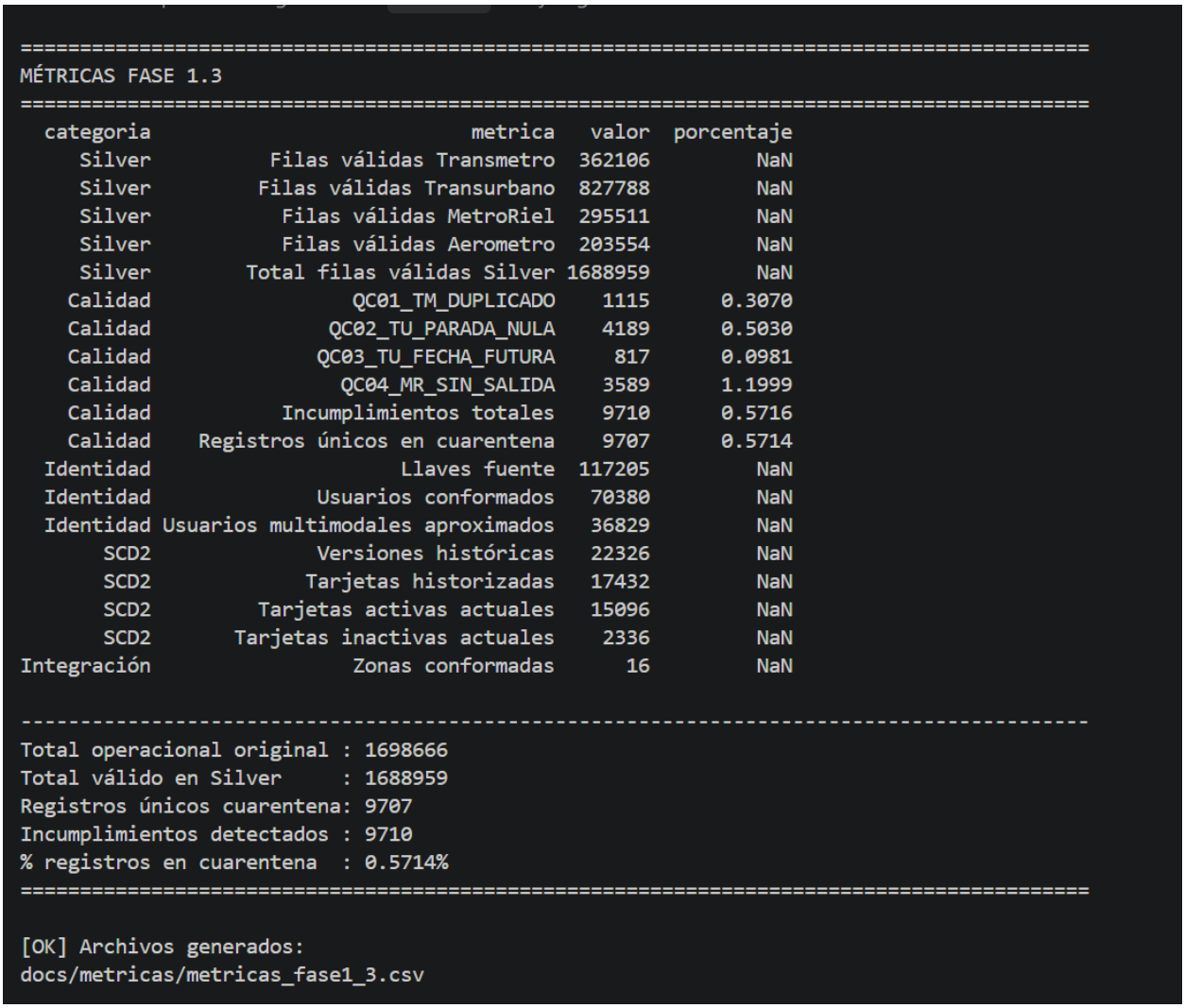
```
(.venv) PS C:\Users\jujar\OneDrive\Escritorio\BackUp\U Landivar\Segundo semestre 2026\Ciencia de datos\Proyecto_Red_Metropolitana> docker exec -it redmetro-postgres psql -U red_user -d red_metropolitana -c "SELECT regla_calidad, COUNT(*) AS cantidad FROM quarantine.registros_invalidos GROUP BY regla_calidad ORDER BY regla_calidad;"
     regla_calidad     | cantidad
-----------------------+----------
 QC01_TM_DUPLICADO     |     1115
 QC02_TU_PARADA_NULA   |     4189
 QC03_TU_FECHA_FUTURA  |      817
 QC04_MR_SIN_SALIDA    |     3589
(4 rows)
```

Figura 19. Pruebas automáticas de calidad de la capa Silver con dbt



```
20:23:43  15 of 26 PASS not_null_silver_metroriel_viajes_trip_id ..... [PASS in 0.50s]
20:23:43  17 of 26 START test not_null_silver_transmetro_validaciones_usuario_id ..... [RUN]
20:23:43  16 of 26 PASS not_null_silver_metroriel_viajes_usuario_id ..... [PASS in 0.70s]
20:23:43  18 of 26 START test not_null_silver_transmetro_validaciones_validacion_id ..... [RUN]
20:23:52  14 of 26 PASS not_null_silver_aerometro_boardings_usuario_id ..... [PASS in 10.16s]
20:23:52  19 of 26 START test not_null_silver_transurbano_transacciones_transaccion_id ... [RUN]
20:23:52  13 of 26 PASS not_null_silver_aerometro_boardings_boarding_id ..... [PASS in 10.25s]
20:23:52  20 of 26 START test not_null_silver_transurbano_transacciones_usuario_id ..... [RUN]
20:23:52  19 of 26 PASS not_null_silver_transurbano_transacciones_transaccion_id ..... [PASS in 0.35s]
20:23:52  21 of 26 START test unique_dim_zona_conformada_zona_id ..... [RUN]
20:23:53  21 of 26 PASS unique_dim_zona_conformada_zona_id ..... [PASS in 0.19s]
20:23:53  22 of 26 START test unique_dim_zona_conformada_zona_nombre ..... [RUN]
20:23:53  20 of 26 PASS not_null_silver_transurbano_transacciones_usuario_id ..... [PASS in 0.61s]
20:23:53  23 of 26 START test unique_silver_aerometro_boardings_boarding_id ..... [RUN]
20:23:53  22 of 26 PASS unique_dim_zona_conformada_zona_nombre ..... [PASS in 0.16s]
20:23:53  24 of 26 START test unique_silver_metroriel_viajes_trip_id ..... [RUN]
20:23:55  17 of 26 PASS not_null_silver_transmetro_validaciones_usuario_id ..... [PASS in 12.13s]
20:23:55  18 of 26 PASS not_null_silver_transmetro_validaciones_validacion_id ..... [PASS in 11.93s]
20:23:55  25 of 26 START test unique_silver_transmetro_validaciones_validacion_id ..... [RUN]
20:23:55  26 of 26 START test unique_silver_transurbano_transacciones_transaccion_id ..... [RUN]
20:23:55  23 of 26 PASS unique_silver_aerometro_boardings_boarding_id ..... [PASS in 2.49s]
20:23:56  24 of 26 PASS unique_silver_metroriel_viajes_trip_id ..... [PASS in 2.79s]
20:23:56  25 of 26 PASS unique_silver_transmetro_validaciones_validacion_id ..... [PASS in 1.38s]
20:24:01  26 of 26 PASS unique_silver_transurbano_transacciones_transaccion_id ..... [PASS in 6.81s]
20:24:02
20:24:02  Finished running 26 data tests in 0 hours 0 minutes and 26.54 seconds (26.54s).
20:24:02
20:24:02  Completed successfully
20:24:02
20:24:02  Done. PASS=26 WARN=0 ERROR=0 SKIP=0 NO-OP=0 REUSED=0 TOTAL=26
```

Figura 20. Métricas finales de la capa Silver y registros de calidad

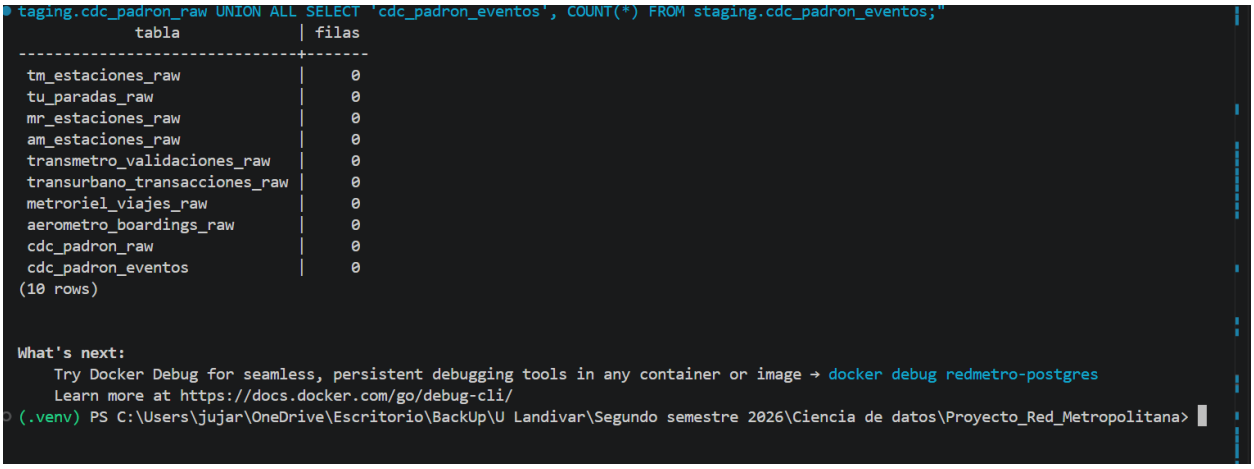


```
================================================================================================
MÉTRICAS FASE 1.3
================================================================================================
  categoria                          metrica   valor  porcentaje
     Silver          Filas válidas Transmetro  362106         NaN
     Silver         Filas válidas Transurbano  827788         NaN
     Silver           Filas válidas MetroRiel  295511         NaN
     Silver           Filas válidas Aerometro  203554         NaN
     Silver        Total filas válidas Silver 1688959         NaN
    Calidad                 QC01_TM_DUPLICADO    1115      0.3070
    Calidad               QC02_TU_PARADA_NULA    4189      0.5030
    Calidad              QC03_TU_FECHA_FUTURA     817      0.0981
    Calidad                QC04_MR_SIN_SALIDA    3589      1.1999
    Calidad           Incumplimientos totales    9710      0.5716
    Calidad    Registros únicos en cuarentena    9707      0.5714
  Identidad                      Llaves fuente  117205         NaN
  Identidad              Usuarios conformados   70380         NaN
  Identidad Usuarios multimodales aproximados   36829         NaN
       SCD2              Versiones históricas   22326         NaN
       SCD2             Tarjetas historizadas   17432         NaN
       SCD2         Tarjetas activas actuales   15096         NaN
       SCD2       Tarjetas inactivas actuales    2336         NaN
Integración                 Zonas conformadas      16         NaN

------------------------------------------------------------------------------------------------
Total operacional original : 1698666
Total válido en Silver     : 1688959
Registros únicos cuarentena: 9707
Incumplimientos detectados : 9710
% registros en cuarentena  : 0.5714%
================================================================================================

[OK] Archivos generados:
docs/metricas/metricas_fase1_3.csv
```

Figura 21. Vaciado de las tablas temporales de Staging después de construir Silver



```
taging.cdc_padron_raw UNION ALL SELECT 'cdc_padron_eventos', COUNT(*) FROM staging.cdc_padron_eventos;"
            tabla            | filas
-----------------------------+-------
 tm_estaciones_raw           |     0
 tu_paradas_raw              |     0
 mr_estaciones_raw           |     0
 am_estaciones_raw           |     0
 transmetro_validaciones_raw |     0
 transurbano_transacciones_raw |     0
 metroriel_viajes_raw        |     0
 aerometro_boardings_raw     |     0
 cdc_padron_raw              |     0
 cdc_padron_eventos          |     0
(10 rows)


What's next:
    Try Docker Debug for seamless, persistent debugging tools in any container or image → docker debug redmetro-postgres
    Learn more at https://docs.docker.com/go/debug-cli/
(.venv) PS C:\Users\jujar\OneDrive\Escritorio\BackUp\U Landivar\Segundo semestre 2026\Ciencia de datos\Proyecto_Red_Metropolitana>
```

Figura 22. Construcción de las dimensiones de la capa Gold con dbt

```
21:52:14  Performance info: dbt\target\perf_info.json
(.venv) PS C:\Users\jujar\OneDrive\Escritorio\BackUp\U Landivar\Segundo semestre 2026\Ciencia de datos\Proyecto_Red_Metropolitana> dbt
run --project-dir dbt --select dim_modo dim_zona dim_usuario dim_tiempo dim_punto_servicio
21:53:48  Running with dbt=1.12.5
21:53:49  Registered adapter: postgres=1.11.0
21:53:53  Found 22 models, 47 data tests, 9 sources, 479 macros
21:53:53
21:53:53  Concurrency: 4 threads (target='dev')
21:53:53
21:53:56  1 of 5 START sql table model gold.dim_modo ..................................... [RUN]
21:53:56  2 of 5 START sql table model gold.dim_tiempo ................................... [RUN]
21:53:56  3 of 5 START sql table model gold.dim_usuario .................................. [RUN]
21:53:56  4 of 5 START sql table model gold.dim_zona ..................................... [RUN]
21:53:57  1 of 5 OK created sql table model gold.dim_modo ................................ [SELECT 4 in 1.09s]
21:53:57  4 of 5 OK created sql table model gold.dim_zona ................................ [SELECT 16 in 1.09s]
21:53:57  5 of 5 START sql table model gold.dim_punto_servicio ........................... [RUN]
21:54:00  2 of 5 OK created sql table model gold.dim_tiempo .............................. [SELECT 64800 in 3.88s]
21:54:07  3 of 5 OK created sql table model gold.dim_usuario ............................. [SELECT 70380 in 11.04s]
21:54:51  5 of 5 OK created sql table model gold.dim_punto_servicio ...................... [SELECT 468 in 53.75s]
21:54:51
21:54:51  Finished running 5 table models in 0 hours 0 minutes and 58.26 seconds (58.26s).
21:54:51
21:54:51  Completed successfully
21:54:51
21:54:51  Done. PASS=5 WARN=0 ERROR=0 SKIP=0 NO-OP=0 REUSED=0 TOTAL=5
(.venv) PS C:\Users\jujar\OneDrive\Escritorio\BackUp\U Landivar\Segundo semestre 2026\Ciencia de datos\Proyecto_Red_Metropolitana>
```

Figura 23. Conteo de registros de las dimensiones Gold

```
(.venv) PS C:\Users\jujar\OneDrive\Escritorio\BackUp\U Landivar\Segundo semestre 2026\Ciencia de datos\Proyecto_Red_Metropolitana> dock
er exec -it redmetro-postgres psql -U red_user -d red_metropolitana -c "SELECT 'dim_modo' AS tabla, COUNT(*) AS filas FROM gold.dim_mod
o UNION ALL SELECT 'dim_zona', COUNT(*) FROM gold.dim_zona UNION ALL SELECT 'dim_usuario', COUNT(*) FROM gold.dim_usuario UNION ALL SEL
ECT 'dim_tiempo', COUNT(*) FROM gold.dim_tiempo UNION ALL SELECT 'dim_punto_servicio', COUNT(*) FROM gold.dim_punto_servicio;"
       tabla        | filas
--------------------+-------
 dim_punto_servicio |   468
 dim_modo           |     4
 dim_zona           |    16
 dim_tiempo         | 64800
 dim_usuario        | 70380
(5 rows)


What's next:
    Try Docker Debug for seamless, persistent debugging tools in any container or image → docker debug redmetro-postgres
    Learn more at https://docs.docker.com/go/debug-cli/
(.venv) PS C:\Users\jujar\OneDrive\Escritorio\BackUp\U Landivar\Segundo semestre 2026\Ciencia de datos\Proyecto_Red_Metropolitana>
```

```
(.venv) PS C:\Users\jujar\OneDrive\Escritorio\BackUp\U Landivar\Segundo semestre 2026\Ciencia de datos\Proyecto_Red_Metropolitana> dbt
run --project-dir dbt --select fact_abordajes fact_viajes_metroriel
23:42:53  Running with dbt=1.12.5
23:42:54  Registered adapter: postgres=1.11.0
23:42:56  Found 24 models, 74 data tests, 9 sources, 479 macros
23:42:56
23:42:56  Concurrency: 4 threads (target='dev')
23:42:56
23:42:57  1 of 2 START sql table model gold.fact_abordajes ............................... [RUN]
23:42:57  2 of 2 START sql table model gold.fact_viajes_metroriel ........................ [RUN]
23:43:32  2 of 2 OK created sql table model gold.fact_viajes_metroriel ................... [SELECT 295511 in 34.39s]
23:45:18  1 of 2 OK created sql table model gold.fact_abordajes .......................... [SELECT 1688959 in 141.00s]
23:45:18
23:45:18  Finished running 2 table models in 0 hours 2 minutes and 22.73 seconds (142.73s).
23:45:19
23:45:19  Completed successfully
23:45:19
23:45:19  Done. PASS=2 WARN=0 ERROR=0 SKIP=0 NO-OP=0 REUSED=0 TOTAL=2
(.venv) PS C:\Users\jujar\OneDrive\Escritorio\BackUp\U Landivar\Segundo semestre 2026\Ciencia de datos\Proyecto_Red_Metropolitana> dock
er exec -it redmetro-postgres psql -U red_user -d red_metropolitana -c "SELECT 'fact_abordajes' AS tabla, COUNT(*) AS filas FROM gold.f
```

Figura 24. Construcción de las tablas de hechos de la capa Gold

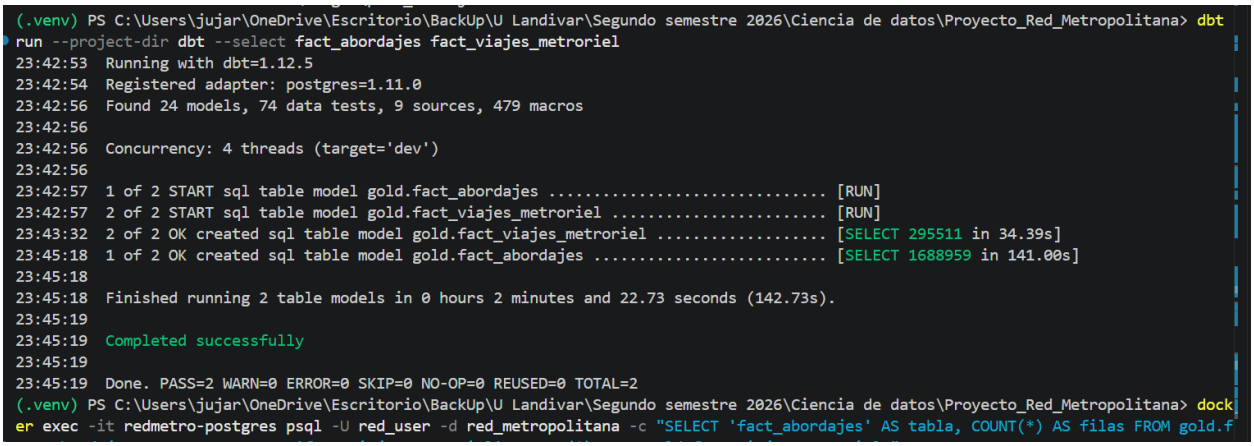


Figura 25. Conteos de las tablas de hechos de Gold

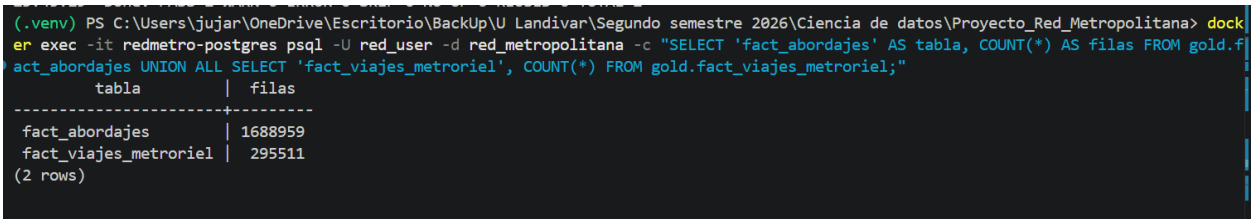


Figura 26. Primera ejecución completa del pipeline mediante Prefect

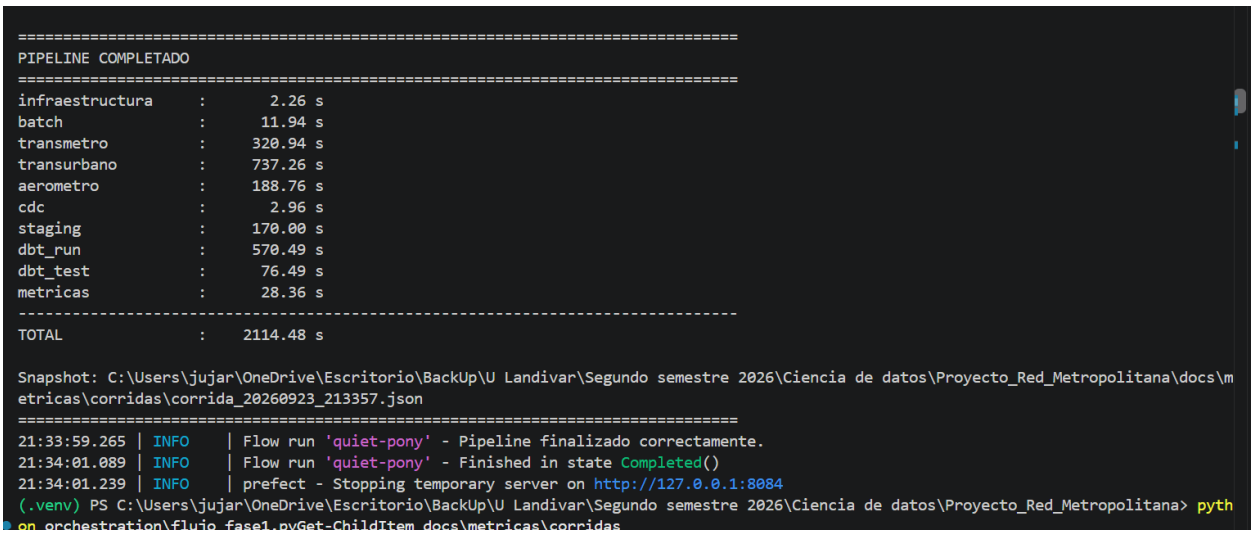
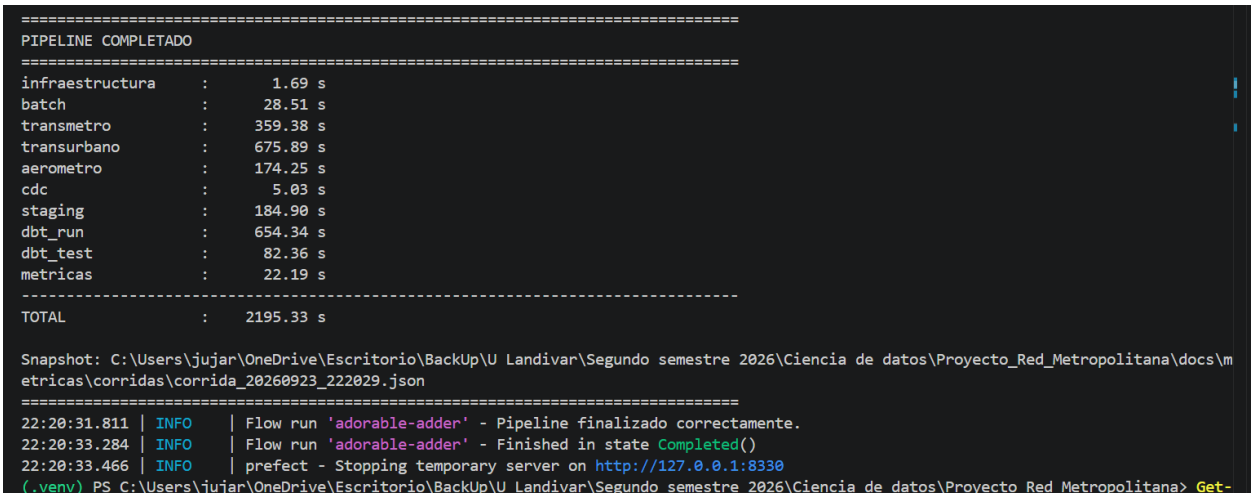


Figura 27. Segunda ejecución completa del pipeline con los mismos datos



```
================================================================================
PIPELINE COMPLETADO
================================================================================
infraestructura     :      1.69 s
batch               :     28.51 s
transmetro          :    359.38 s
transurbano         :    675.89 s
aerometro           :    174.25 s
cdc                 :      5.03 s
staging             :    184.90 s
dbt_run             :    654.34 s
dbt_test            :     82.36 s
metricas            :     22.19 s
--------------------------------------------------------------------------------
TOTAL               :   2195.33 s

Snapshot: C:\Users\jujar\OneDrive\Escritorio\BackUp\U Landivar\Segundo semestre 2026\Ciencia de datos\Proyecto_Red_Metropolitana\docs\m
etricas\corridas\corrida_20260923_222029.json
================================================================================
22:20:31.811 | INFO    | Flow run 'adorable-adder' - Pipeline finalizado correctamente.
22:20:33.284 | INFO    | Flow run 'adorable-adder' - Finished in state Completed()
22:20:33.466 | INFO    | prefect - Stopping temporary server on http://127.0.0.1:8330
(.venv) PS C:\Users\jujar\OneDrive\Escritorio\BackUp\U Landivar\Segundo semestre 2026\Ciencia de datos\Proyecto_Red_Metropolitana> Get-
```

Figura 28. Verificación de idempotencia entre dos ejecuciones del pipeline

```
Corrida 1: corrida_20260923_213357.json
Corrida 2: corrida_20260923_222029.json

   capa                                objeto  corrida_1  corrida_2  igual
 Bronze                             aerometro     203554     203554   True
 Bronze                          am_estaciones         14         14   True
 Bronze                             cdc_padron      31050      31050   True
 Bronze                              metroriel     299100     299100   True
 Bronze                          mr_estaciones         22         22   True
 Bronze                          tm_estaciones        104        104   True
 Bronze                             transmetro     363221     363221   True
 Bronze                           transurbano     832791     832791   True
 Bronze                            tu_paradas        328        328   True
 Silver       silver.bridge_identidad_usuarios     117205     117205   True
 Silver             silver.dim_zona_conformada         16         16   True
 Silver          silver.scd2_padron_transmetro      22326      22326   True
 Silver      silver.silver_aerometro_boardings     203554     203554   True
 Silver         silver.silver_metroriel_viajes     295511     295511   True
 Silver  silver.silver_transmetro_validaciones     362106     362106   True
 Silver silver.silver_transurbano_transacciones     827788     827788   True
   Gold                        gold.dim_modo          4          4   True
   Gold                gold.dim_punto_servicio        468        468   True
   Gold                        gold.dim_tiempo      64800      64800   True
   Gold                       gold.dim_usuario      70380      70380   True
   Gold                          gold.dim_zona         16         16   True
   Gold                   gold.fact_abordajes    1688959    1688959   True
   Gold             gold.fact_viajes_metroriel     295511     295511   True
Calidad                          cuarentena       9710       9710   True

--------------------------------------------------------------------------------------------
[OK] IDEMPOTENCIA VERIFICADA
Los conteos de Bronze, Silver y Gold no cambiaron.
============================================================================================
```